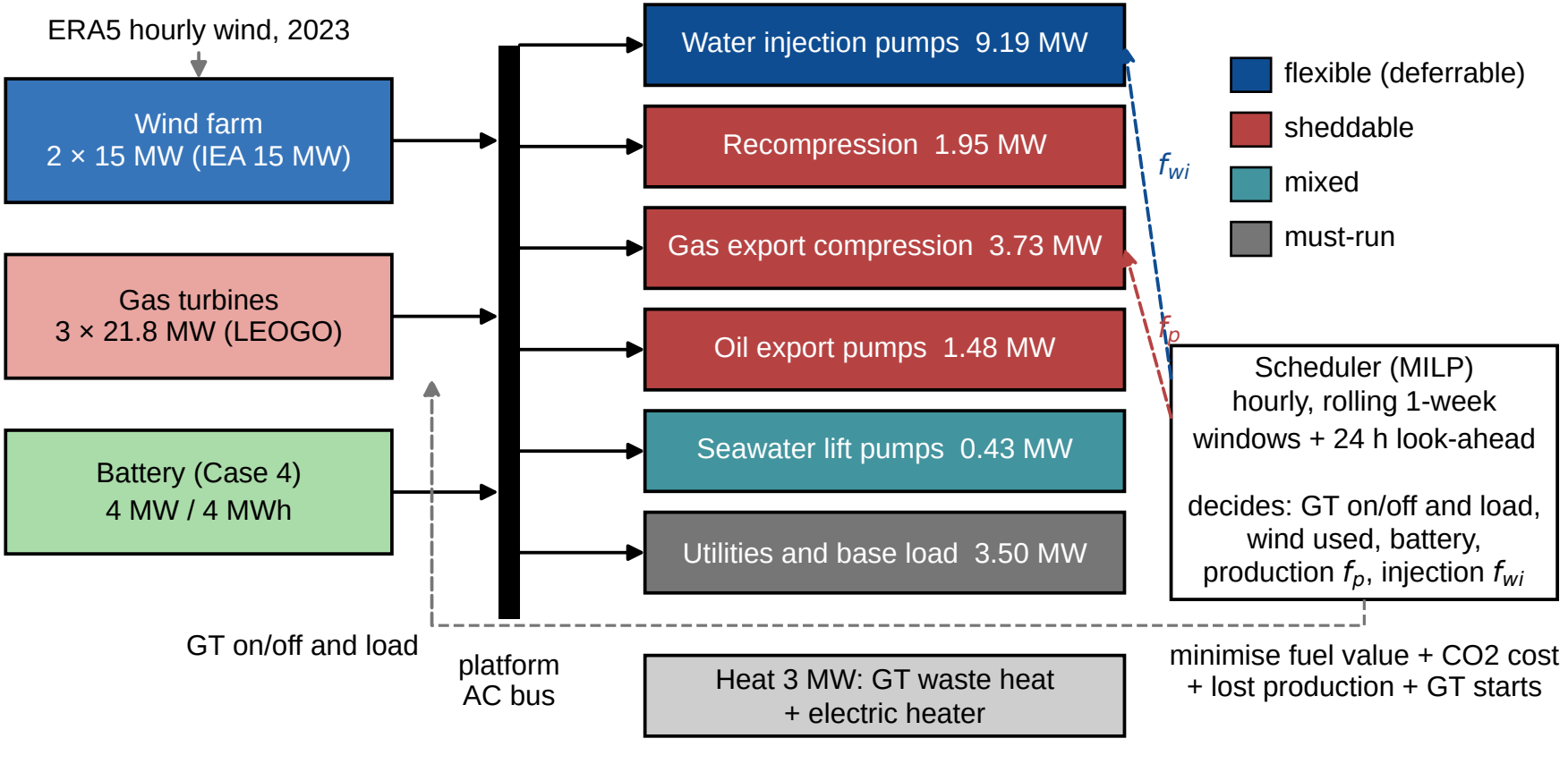
ERA5 hourly wind, 2023
Wind farm
2 × 15 MW (IEA 15 MW)
Gas turbines
3 × 21.8 MW (LEOGO)
Battery (Case 4)
4 MW / 4 MWh
Water injection pumps 9.19 MW
Recompression 1.95 MW
Gas export compression 3.73 MW
Oil export pumps 1.48 MW
Seawater lift pumps 0.43 MW
Utilities and base load 3.50 MW
Heat 3 MW: GT waste heat
+ electric heater
platform
AC bus
GT on/off and load
f_{wi}
f_p
flexible (deferrable)
sheddable
mixed
must-run
Scheduler (MILP)
hourly, rolling 1-week
windows + 24 h look-ahead
decides: GT on/off and load,
wind used, battery,
production f_p, injection f_{wi}
minimise fuel value + CO2 cost
+ lost production + GT starts